SMART MEDICATION SYSTEM

Java Swing Desktop Application using Data Structures

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PROBLEM STATEMENT

EXISTING PROBLEMS

- People often forget to take medicines
- No proper way to manage medicine inventory
- No reminder mechanism
- Difficulty in tracking expiry dates
- Low stock goes unnoticed
- No record of past activities

PROPOSED SOLUTION

Smart Medication System

A desktop application that efficiently manages medicines, reminds users, and helps maintain a healthy life.

OBJECTIVES

01 Add and Manage Medicines

02 Maintain Medicine Inventory

03 Provide Medicine Reminders

04 Monitor Low Stock Medicines

05 Track Medicines Nearing Expiry

06 Maintain Activity History

07 Enable Fast Search

08 Demonstrate Practical Use of Data Structures

02

PROBLEM STATEMENT

The challenges in managing medicines manually

EXISTING PROBLEMS

People often forget to take medicines
No timely reminders lead to missed doses and health risks.

No proper way to manage inventory
Hard to keep track of available medicines and quantities.

No reminder mechanism
Patients may miss important medicine schedules.

Difficulty in tracking expiry dates
Expired medicines can be consumed unknowingly.

Low stock goes unnoticed
Medicines run out without any warning.

No record of past activities
Difficult to review past actions like add, delete or restore.

PROPOSED SOLUTION

Smart Medication System
A desktop application that efficiently manages medicines, reminds users, and helps maintain a healthy life.

Our system automates medicine management, **reduces human error,** and ensures **timely intake for better health.**

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OBJECTIVES

What this system aims to achieve

01 Add & Manage Medicines

Allow users to add and manage medicines with all necessary details.

02 Maintain Medicine Inventory

Keep an organized list of all medicines with real-time updates.

03 Provide Medicine Reminders

Remind users to take medicines on time for better adherence.

04 Monitor Expiry Dates

Track expiry dates and alert users before medicines expire.

05 Track Low Stock Medicines

Alert users when the quantity of any medicine is running low.

06 Maintain Activity History

Store and view history of all actions like add, delete and restore.

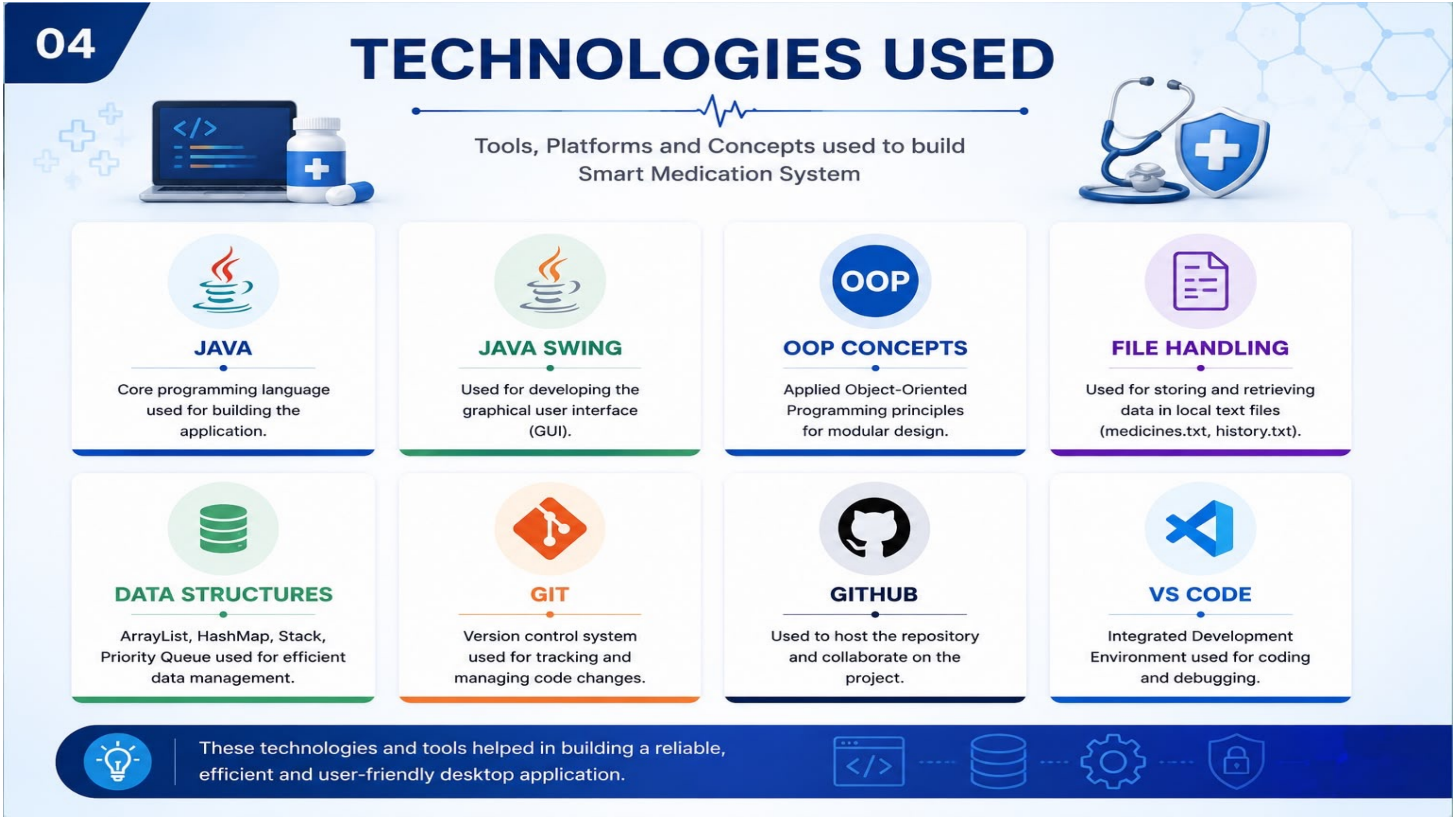
07 Enable Fast Search

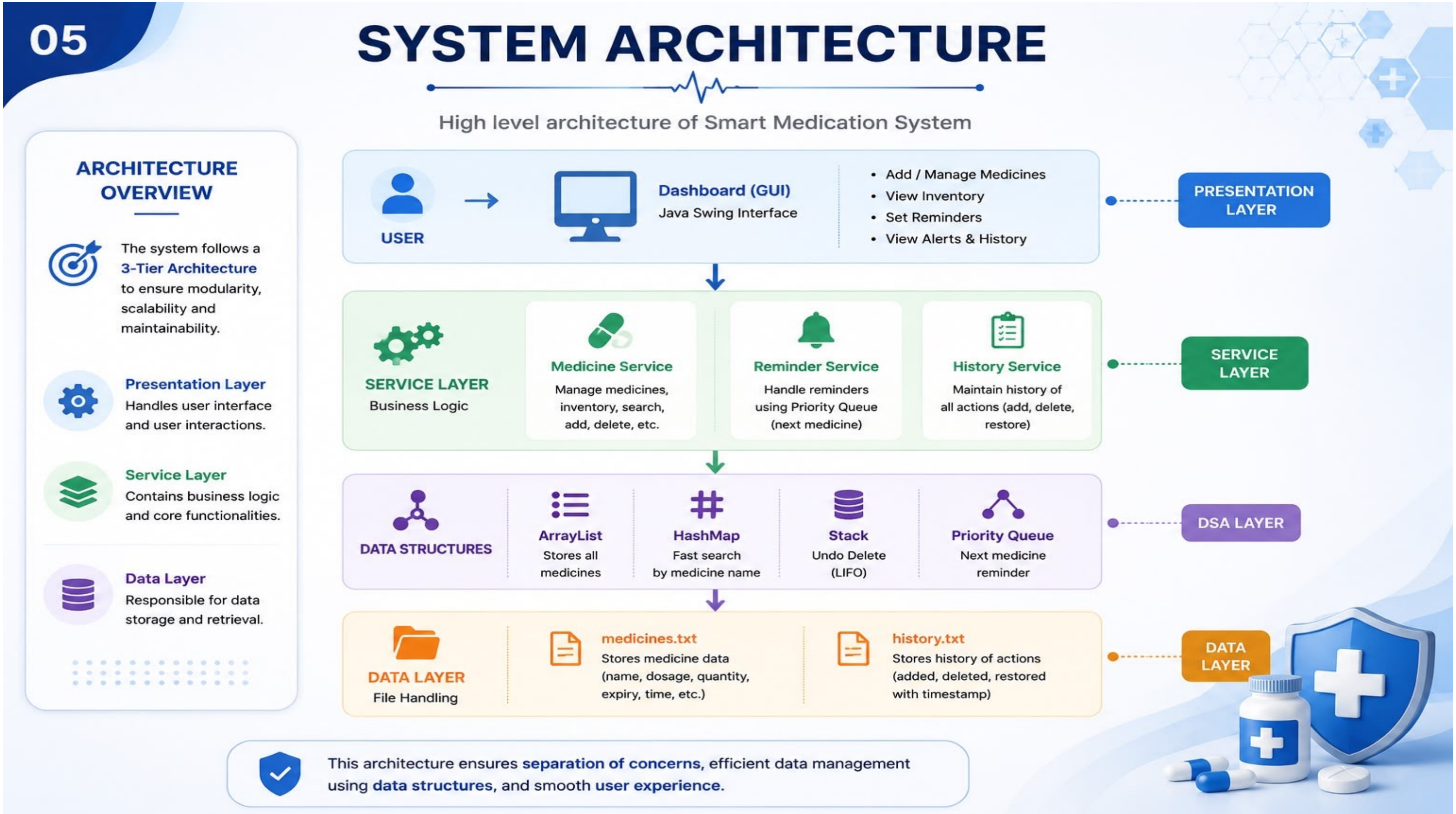
Search medicines quickly using optimized data structures.

08 Demonstrate DSA Concepts

Apply data structures effectively to solve real-world problems.

Goal: To build a reliable, efficient and user-friendly system that helps individuals manage their medicines and improve their health.


04
TECHNOLOGIES USED
Tools, Platforms and Concepts used to build Smart Medication System
JAVA
Core programming language used for building the application.
JAVA SWING
Used for developing the graphical user interface (GUI).
OOP
OOP CONCEPTS
Applied Object-Oriented Programming principles for modular design.
FILE HANDLING
Used for storing and retrieving data in local text files (medicines.txt, history.txt).
DATA STRUCTURES
ArrayList, HashMap, Stack, Priority Queue used for efficient data management.
GIT
Version control system used for tracking and managing code changes.
GITHUB
Used to host the repository and collaborate on the project.
VS CODE
Integrated Development Environment used for coding and debugging.
These technologies and tools helped in building a reliable, efficient and user-friendly desktop application.


05
SYSTEM ARCHITECTURE
High level architecture of Smart Medication System
ARCHITECTURE OVERVIEW
The system follows a 3-Tier Architecture to ensure modularity, scalability and maintainability.
Presentation Layer
Handles user interface and user interactions.
Service Layer
Contains business logic and core functionalities.
Data Layer
Responsible for data storage and retrieval.
USER
Dashboard (GUI)
Java Swing Interface
• Add / Manage Medicines
• View Inventory
• Set Reminders
• View Alerts & History
PRESENTATION LAYER
SERVICE LAYER
Business Logic
Medicine Service
Manage medicines, inventory, search, add, delete, etc.
Reminder Service
Handle reminders using Priority Queue (next medicine)
History Service
Maintain history of all actions (add, delete, restore)
SERVICE LAYER
DATA STRUCTURES
ArrayList
Stores all medicines
HashMap
Fast search by medicine name
Stack
Undo Delete (LIFO)
Priority Queue
Next medicine reminder
DSA LAYER
DATA LAYER
File Handling
medicines.txt
Stores medicine data (name, dosage, quantity, expiry, time, etc.)
history.txt
Stores history of actions (added, deleted, restored with timestamp)
DATA LAYER
This architecture ensures separation of concerns, efficient data management using data structures, and smooth user experience.

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DATA STRUCTURES USED

The backbone of Smart Medication System

WHY DIFFERENT DATA STRUCTURES?

Each data structure is chosen based on the nature of the operation to ensure optimal performance and efficiency.

01 ARRAYLIST

Medicine Storage

Used to store all medicines. Supports dynamic size and easy iteration.

EXAMPLE

List<Medicine> medicines

COMPLEXITY

Add : O(1)
Search : O(n)
Delete : O(n)

02 HASHMAP

Fast Search

KEY	VALUE

Used to search medicines quickly by name. Provides O(1) average complexity.

EXAMPLE

Map<String, Medicine> medicineMap

COMPLEXITY

Search : O(1)
Insert : O(1)
Delete : O(1)

03 STACK

Undo Delete

Stores deleted medicines to support Undo operation. Follows LIFO principle.

EXAMPLE

Stack<Medicine> undoStack

COMPLEXITY

Push : O(1)
Pop : O(1)
Peek : O(1)

04 PRIORITY QUEUE

Medicine Reminder

Stores medicines based on reminder time. The earliest medicine is retrieved first.

EXAMPLE

PriorityQueue<Reminder> reminderQueue

COMPLEXITY

Insert : O(log n)
Remove : O(log n)
Peek : O(1)

IMPACT: Using the right data structures improved performance, reduced time complexity and provided a smooth and efficient user experience.

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FEATURES IMPLEMENTED

Smart Medication System – All in One Solution

01 ADD MEDICINE
Add new medicines with details like name, dosage, time, quantity and expiry date.

02 INVENTORY
View all medicines in inventory with complete details in a structured manner.

03 SEARCH MEDICINE
Search medicines quickly by name using HashMap for fast and efficient search.

04 DELETE MEDICINE
Remove medicines from inventory when they are no longer required.

05 UNDO DELETE
Restore the last deleted medicine using Stack (LIFO principle).

06 NEXT MEDICINE REMINDER
Get reminder of the next medicine to take based on upcoming time using Priority Queue.

07 LOW STOCK ALERT
Alert when any medicine quantity is running low to avoid unavailability.

08 EXPIRY ALERT
Notify about medicines that are about to expire based on expiry date.

09 HISTORY
Maintain history of all actions like added, deleted and restored medicines.

These features work together to ensure better medicine management, timely reminders and improved health care.

REMIND | SAFE | ON TIME | HEALTHY

08

DASHBOARD (HOME SCREEN)

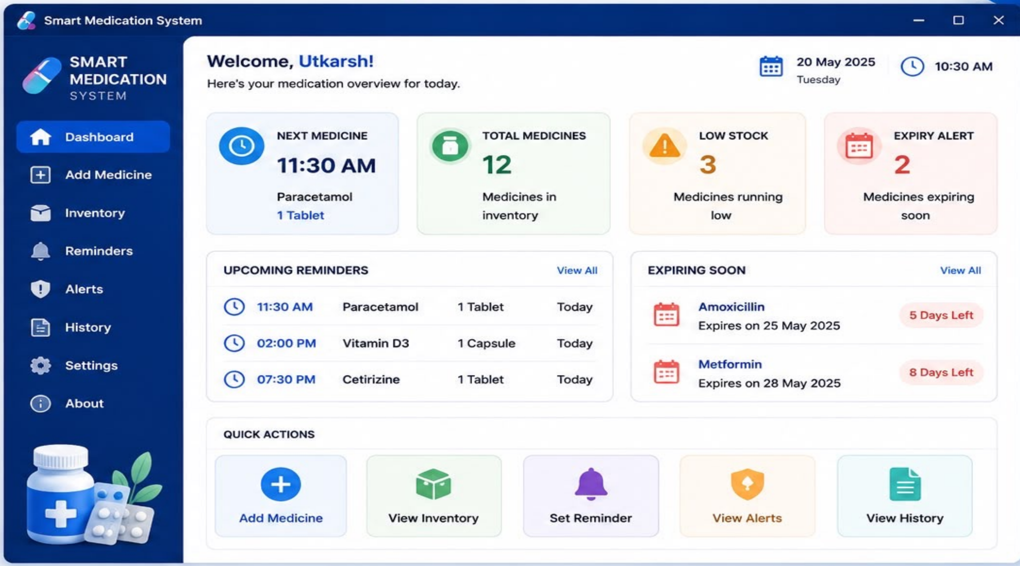
The central hub of Smart Medication System

OVERVIEW

The Dashboard provides a quick overview of your medication schedule, alerts, and inventory status with easy access to all major features.

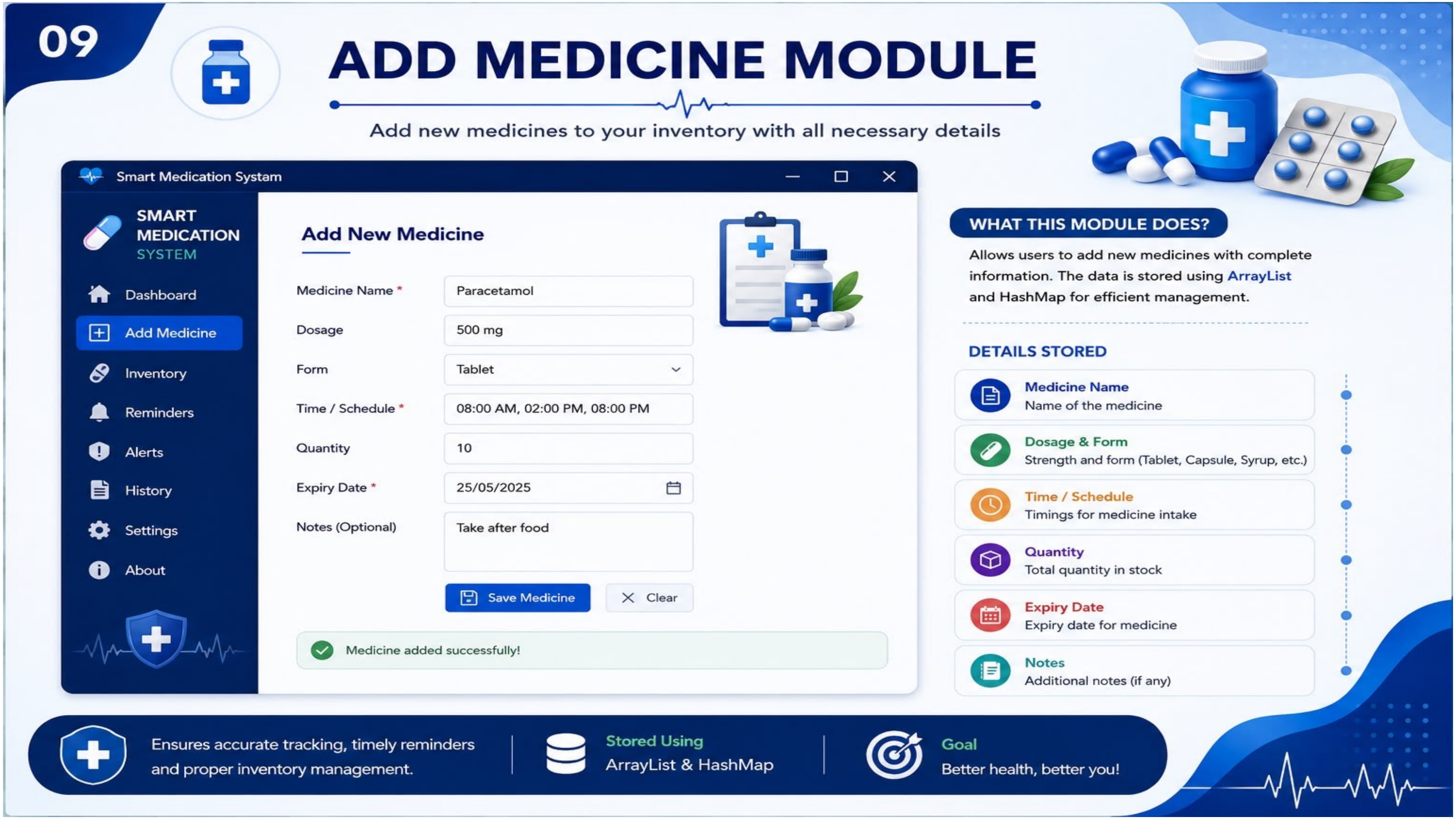
KEY HIGHLIGHTS

- Next medicine reminder at a glance
- Low stock and expiry alerts
- Quick access to all modules
- Clean and modern user interface



Your health, Our priority.

Stay on track. Stay healthy.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ADD MEDICINE MODULE
Add new medicines to your inventory with all necessary details
Smart Medication System
SMART MEDICATION SYSTEM
Dashboard
Add Medicine
Inventory
Reminders
Alerts
History
Settings
About
Add New Medicine
Medicine Name *
Paracetamol
Dosage
500 mg
Form
Tablet
Time / Schedule *
08:00 AM, 02:00 PM, 08:00 PM
Quantity
10
Expiry Date *
25/05/2025
Notes (Optional)
Take after food
Save Medicine
Clear
Medicine added successfully!
WHAT THIS MODULE DOES?
Allows users to add new medicines with complete information. The data is stored using ArrayList and HashMap for efficient management.
DETAILS STORED
Medicine Name
Name of the medicine
Dosage & Form
Strength and form (Tablet, Capsule, Syrup, etc.)
Time / Schedule
Timings for medicine intake
Quantity
Total quantity in stock
Expiry Date
Expiry date for medicine
Notes
Additional notes (if any)
Ensures accurate tracking, timely reminders and proper inventory management.
Stored Using
ArrayList & HashMap
Goal
Better health, better you!

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CONCLUSION & FUTURE SCOPE

Building a Smarter, Healthier Tomorrow

CONCLUSION

The **Smart Medication System** efficiently manages medicines, reminders, and inventory using the power of **Data Structures** and modern technologies. It ensures timely medication, reduces human error, and promotes **better health management**.

- Efficient data handling using ArrayList, HashMap, Stack and Priority Queue
- Fast search, quick reminders, and data reliability
- User-friendly interface for seamless experience
- Scalable, modular and easy to maintain

FUTURE SCOPE

Mobile App Integration	Develop a cross-platform mobile app for better accessibility and real-time alerts.
AI-Powered Reminders	Use machine learning to predict and personalize medication schedules.
Cloud Synchronization	Enable cloud storage for multi-device access and data backup.
Wearable Device Integration	Sync with smartwatches and health devices for real-time health monitoring.
Advanced Analytics	Generate health reports and insights to track medication adherence.

"Technology is best when it brings
CARE, COMFORT & CONTROL
to everyday life."

THANK YOU!

Stay Healthy, Stay Happy

Smart Management Better Health	**Timely Reminders** Never Miss a Dose	**Efficient & Reliable** Powered by DSA	**User-Centric** Designed for You	**Health First** Always

End of Presentation

THANK YOU!

Smart Medication System

Smarter Management, Better Health.

> "Technology is best when it brings care, comfort and control to everyday life."

- Timely Reminders
- Safe & Secure Management
- Better Health Outcomes

ACKNOWLEDGEMENT

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Better Management Today, **Healthier Tomorrow!**

- Efficient Medication Management
- Saves Time & Reduces Errors
- User-Friendly & Accessible Anywhere
- Secure, Reliable & Scalable Solution

Stay on track. Stay healthy. Stay happy.